UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

# Flowcharts and visuals

Procedures, decisions, concept maps and comparison tables.

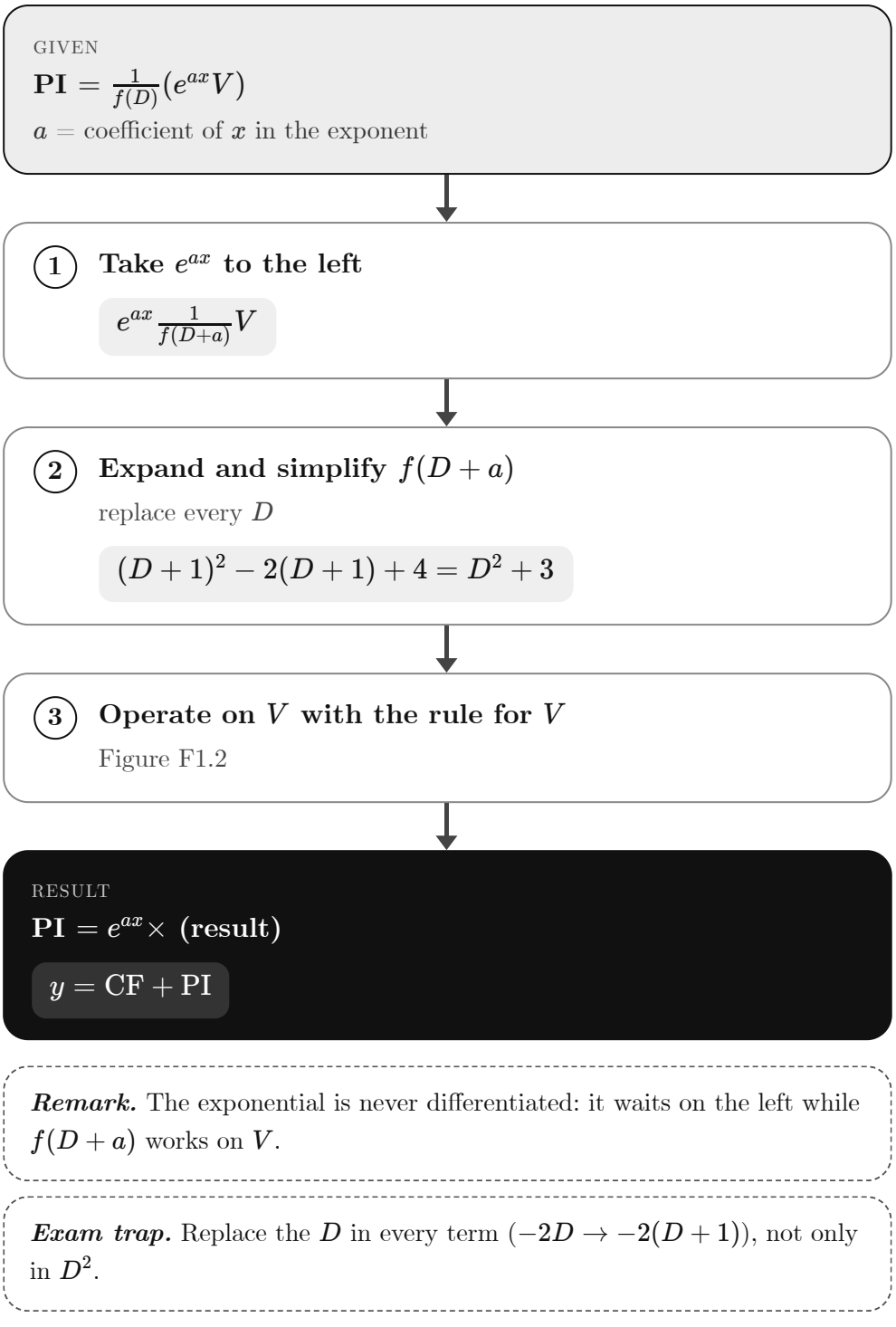
Class 006 · Particular integral Type 4 ($e^{ax}V$) and Type 5 ($xV$) · compiled 16 October 2026

| | |
|---|---|
| ESA ★★★ | Asked repeatedly in past ESA papers |
| ESA ★★ | Asked in past ESA papers or computed in class-work problems |
| ESA ★ | Conceptual: taught in class, not examined so far |
| ADDED FOR CLARITY | Standard context that is not in the class material |
| TRANSCRIPT ONLY | Said in the lecture but not on any slide |
| VIDEO ONLY | Seen only in a frame of the class video |
| NOT ON CURRENT SLIDES | From older material or past papers, not this year's slides |
| S006·5 | Unit 2 deck 006 (“Particular integral of Type 4”), slide 5 (rendered .pptx) |
| T006 00:04:12 | Unit 2 AV-summary transcript “Particular integral of standard functions – Type 4” (17 min), segment start |
| T006b 00:12:47 | Unit 2 AV-summary transcript “Particular integral – Type 3 and Type 4” (bucket 006, 15.5 min); only its Type 4 part (00:09:55–00:15:25) is cited here. The class 005 pack cites the same file's Type 3 part as T006a. |
| CW U2 P18 | Unit 2 class-work problem 18 |
| ESA Jan-24 s3 2c | Past paper: UE23MA141A Jan 2024 set 3, question 2.c |

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4. Figure F2.2 · Slides 006·7–9: $(D-1)^2y = xe^x\cos x$, corrected
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# 1 The Type 4 rule: $X = e^{ax}V$ (exponential shift)

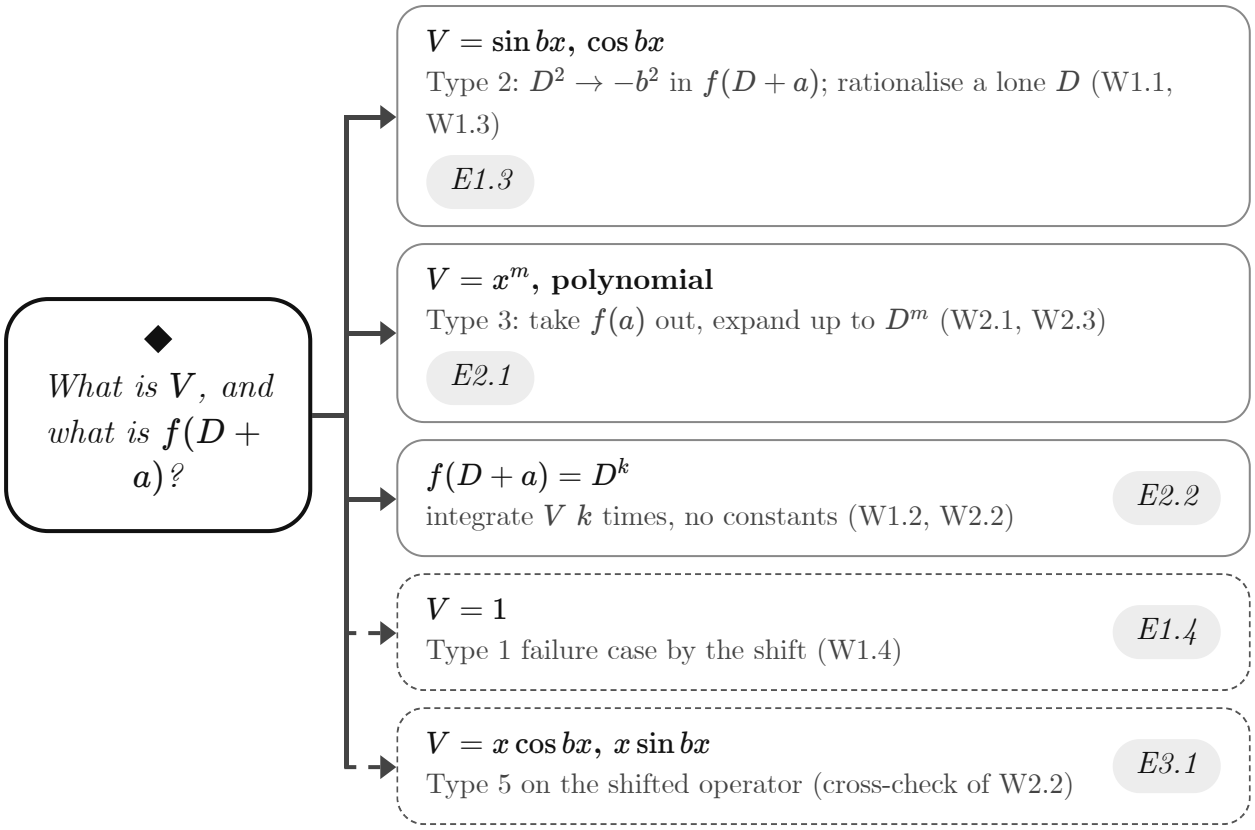



***Remark.*** The exponential is never differentiated: it waits on the left while $f(D + a)$ works on $V$.

***Exam trap.*** Replace the $D$ in every term ($-2D \to -2(D + 1)$), not only in $D^2$.

**Figure F1.1.** The Type 4 procedure (slide 006·4) ESA ★★

S006·4 · S006·10 · T006 00:01:00 · T006b 00:10:19 · T006b 00:10:53




***Remark.*** T006b 00:10:53: "If $V$ is sine or cos, we go with case 2. If $V$ is a polynomial, we go with case 3."

**Figure F1.2.** After the shift: which rule for $V$? ESA ★★

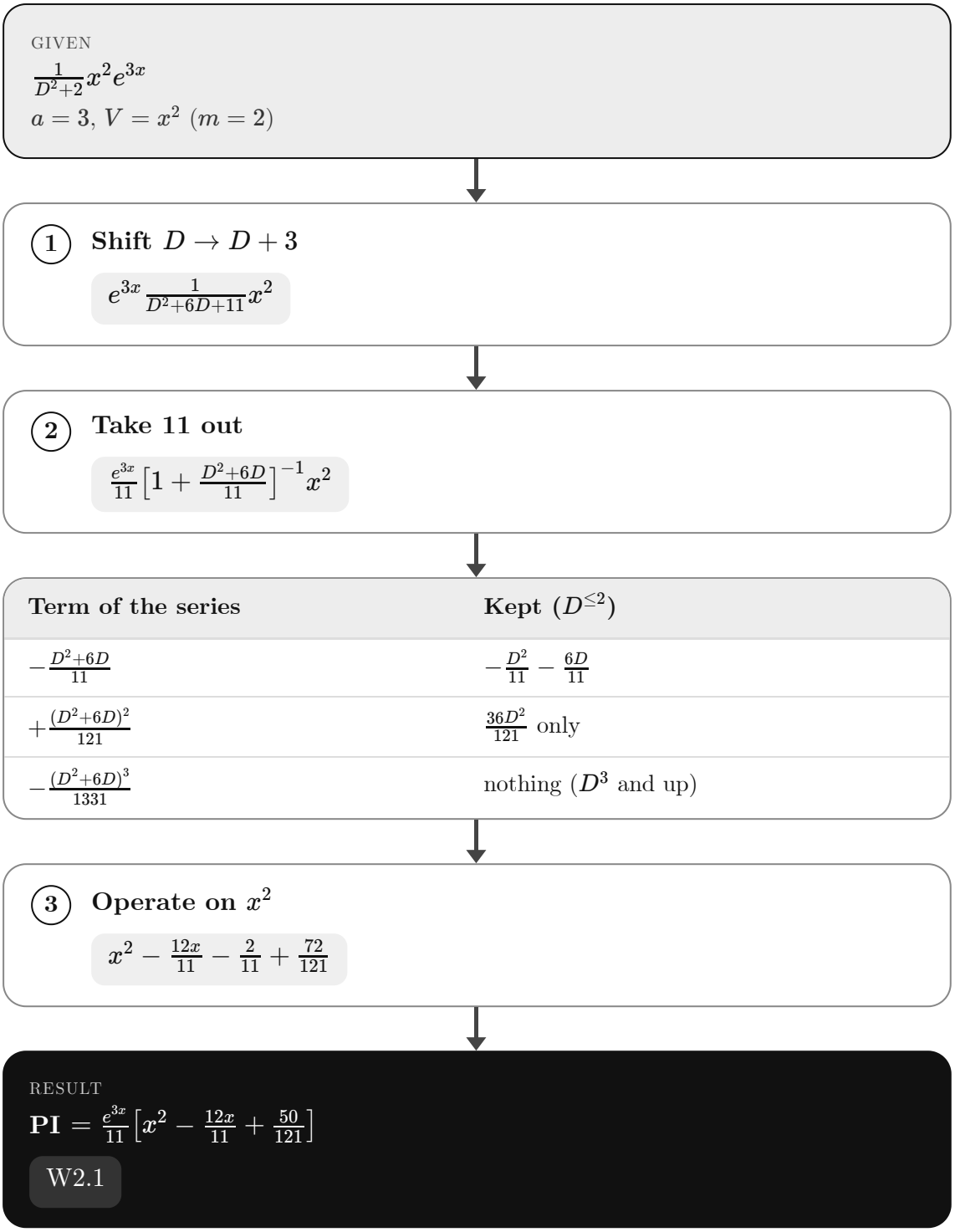
T006b 00:10:53 · S006·6 · S006·8 · S006·12 · S006·14

## The five PI types side by side

| Type | $X$ | Rule | Special case | Class pack |
|---|---|---|---|---|
| **1** | $e^{ax}$ | $D \to a$ | $f(a) = 0$: $x\frac{e^{ax}}{f'(a)}$ | 003 |
| **2** | $\sin ax$, $\cos ax$ | $D^2 \to -a^2$ | $f(-a^2) = 0$: multiply by $x$, differentiate | 004 |
| **3** | $x^m$ | $[f(D)]^{-1}$ in powers of $D$, up to $D^m$ | $f(0) = 0$: take $D^k$ out | 005 |
| **4** | $e^{ax}V$ | $e^{ax}\frac{1}{f(D+a)}V$ | none needed: the shift gives $D^k$ | 006 (this pack) |
| **5** | $xV$ | $\left[\frac{x}{f(D)} - \frac{f'(D)}{(f(D))^2}\right]V$ | use Type 2 on both terms | 006 (this pack) |

ADDED FOR CLARITY S006·4 · S006·10

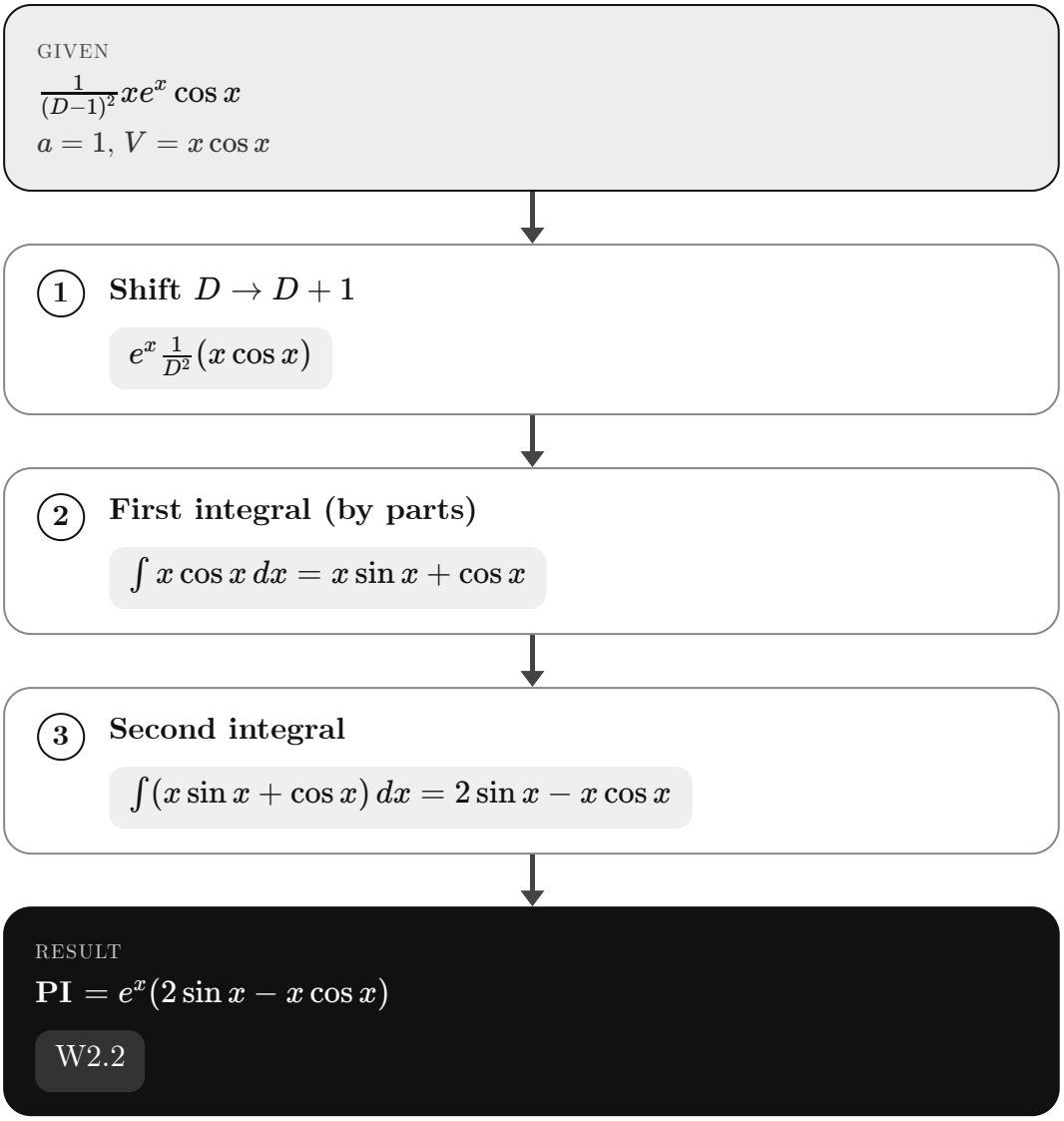
# 2 Type 4 with a polynomial: $x^m e^{ax}$ and mixed right-hand sides




***Remark.*** "$a^2$ is not needed, $b^2$ is needed, $2ab$ is not needed" (T006b 00:13:25).

**Figure F2.1.** Slides 006·11–12: $(D^2+2)y = x^2e^{3x}$ ESA ★★

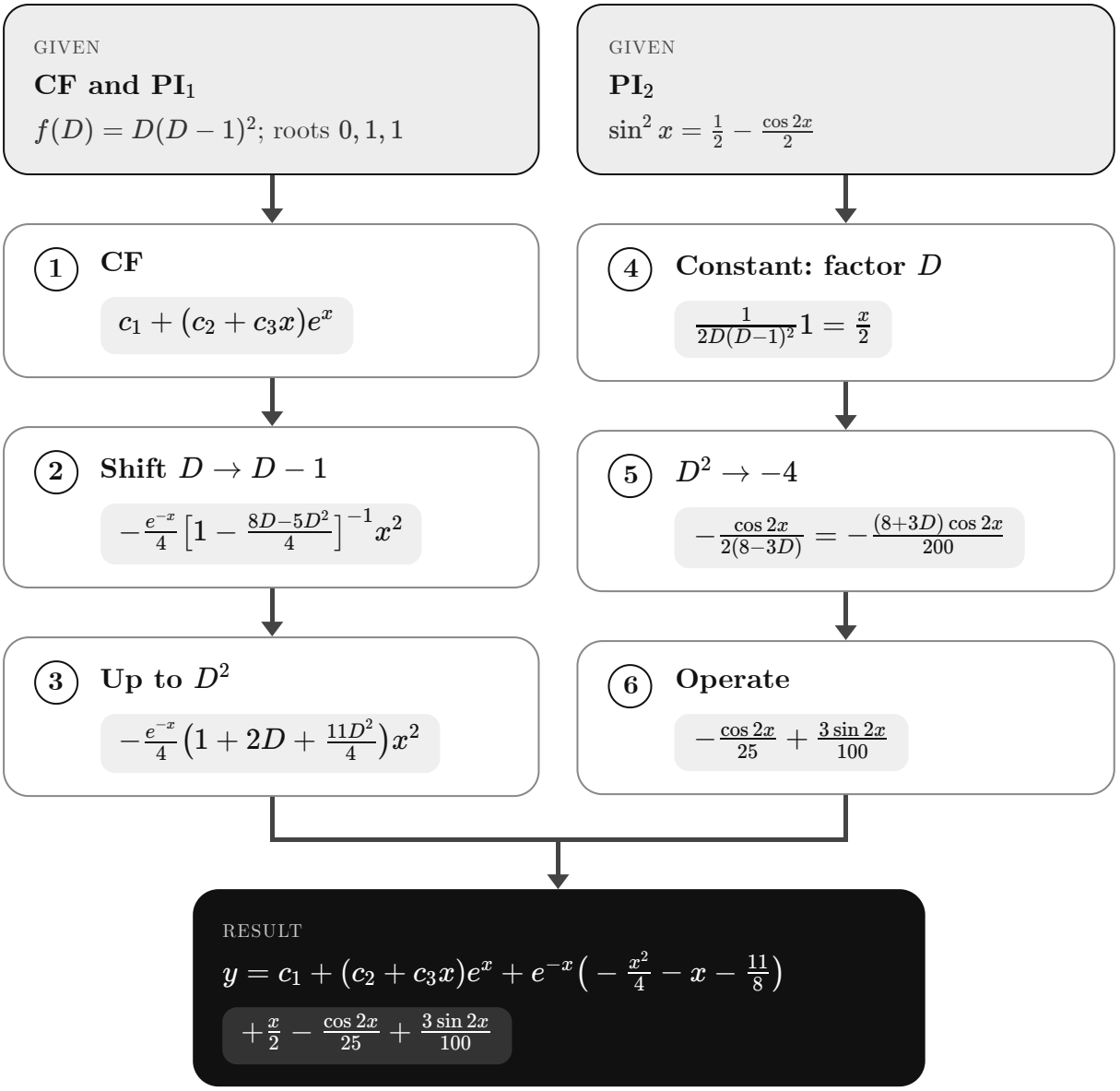
S006·11 · S006·12 · T006b 00:12:13 · T006b 00:13:25




***Exam trap.*** Slide 006·9 integrates $x \sin x$ at step 2 and ends with $-e^x(x \sin x + 2 \cos x)$, which does not satisfy the equation.

**Figure F2.2.** Slides 006·7–9: $(D-1)^2 y = xe^x \cos x$, corrected ESA ★★

S006·8 · S006·9 · T006 00:06:40 · T006 00:07:16




***Remark.*** Worked in full as W2.3 (6 marks). The $\sin^2 x$ part is the class 004 pack's W3.2.

**Figure F2.3.** ESA Jan-24 s3 2c (= CW P18): $(D^3 - 2D^2 + D)y = x^2 e^{-x} + \sin^2 x$
ESA ★★

ESA Jan-24 s3 2c · CW U2 P18

## Every Type 4 and Type 5 problem in this pack

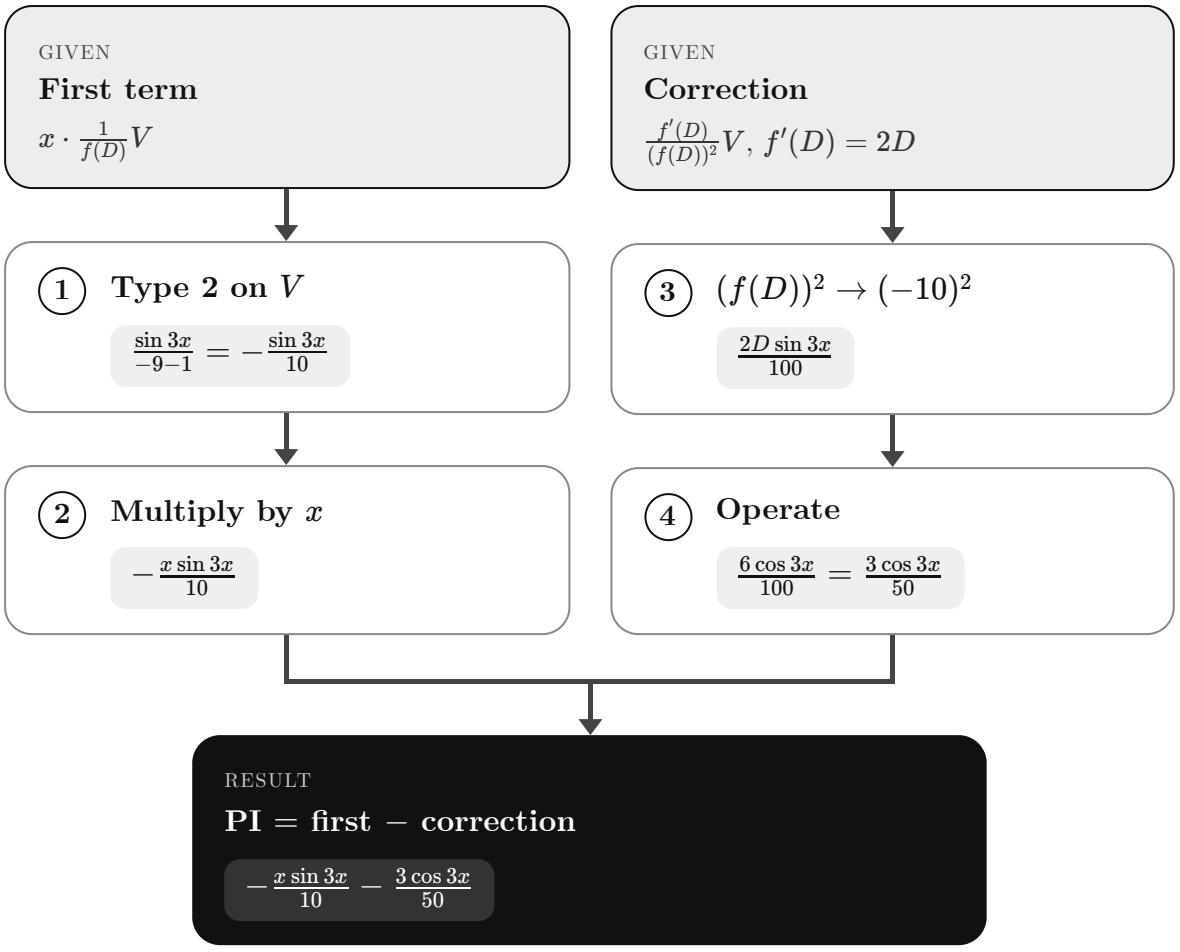
| Source (worked) | Equation | After the shift / rule | PI |
|---|---|---|---|
| **S006·5–6 (W1.1)** | $(D^2 - 2D + 4)y = e^x \cos x$ | $e^x \frac{\cos x}{D^2+3}$ | $\frac{e^x \cos x}{2}$ |
| **S006·13–14 (W1.2)** | $(D-1)^2 y = e^x \sin x$ | $e^x \frac{\sin x}{D^2}$ | $-e^x \sin x$ |

| Source (worked) | Equation | After the shift / rule | PI |
|---|---|---|---|
| **CW P17 (W1.3)** | $(D^2+4D+3)y = e^{2x}\cos x$ | $e^{2x}\frac{\cos x}{D^2+8D+15}$ | $\frac{e^{2x}(7\cos x+4\sin x)}{130}$ |
| **S006·11–12 (W2.1)** | $(D^2+2)y = x^2e^{3x}$ | $e^{3x}\frac{x^2}{D^2+6D+11}$ | $\frac{e^{3x}}{11}(x^2-\frac{12x}{11}+\frac{50}{121})$ |
| **S006·7–9 (W2.2)** | $(D-1)^2y = xe^x\cos x$ | $e^x\frac{x\cos x}{D^2}$ | $e^x(2\sin x - x\cos x)$ |
| **ESA s3 2c = P18 (W2.3)** | $(D^3-2D^2+D)y = x^2e^{-x}+\dots$ | $e^{-x}\frac{x^2}{D^3-5D^2+8D-4}$ | $e^{-x}(-\frac{x^2}{4}-x-\frac{11}{8})+\dots$ |
| **T006 only (W2.4)** | $(D^2-3D+2)y = xe^{3x}+\sin 2x$ | $e^{3x}\frac{x}{D^2+3D+2}$ | $\frac{e^{3x}}{2}(x-\frac{3}{2})+\frac{3\cos 2x-\sin 2x}{20}$ |
| **CW P19 (W3.1)** | $(D^2-1)y = x\sin 3x$ | Type 5 | $-\frac{x\sin 3x}{10}-\frac{3\cos 3x}{50}$ |
| **CW P20 (W3.2)** | $(D^2-5D+6)y = x\cos 2x$ | Type 5 | $\frac{x(\cos 2x-5\sin 2x)}{52}-\frac{73\sin 2x}{1352}-\frac{5\cos 2x}{169}$ |

ADDED FOR CLARITY

S006·5 · S006·7 · S006·11 · S006·13 · T006 00:09:45 · CW U2 P17 · CW U2 P18 · CW U2 P19 · CW U2 P20 · ESA Jan-24 s3 2c

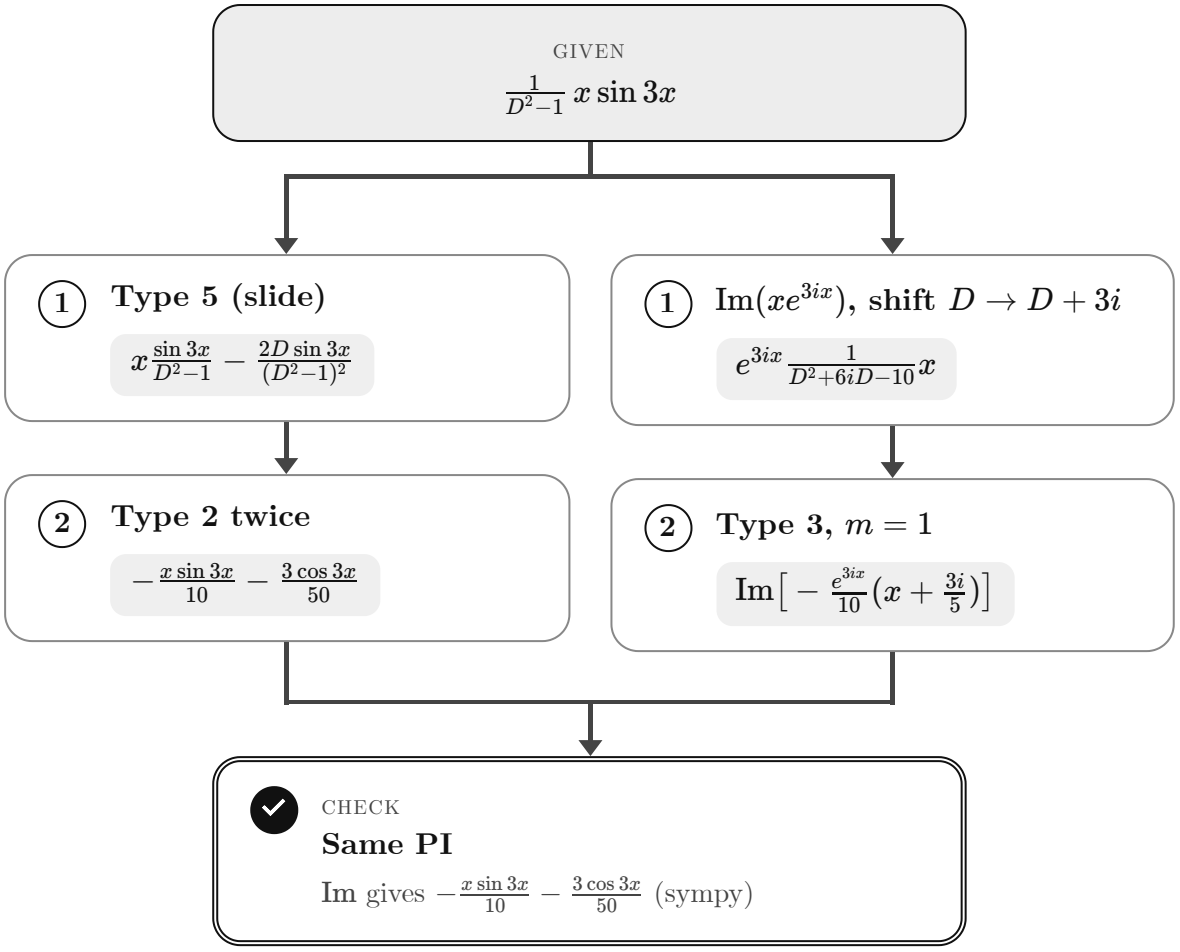
# 3 The Type 5 rule: $X = xV$




***Remark.*** Both halves are ordinary Type 2 work. Worked as W3.1; W3.2 (CW P20) needs rationalising in both halves.

**Figure F3.1.** The Type 5 procedure, on CW P19: $(D^2 - 1)y = x \sin 3x$ ESA ★★

S006·4 · CW U2 P19




***Remark.*** The right-hand route is added for clarity; write the slide method (left) in the exam.

**Figure F3.2.** Two routes to the PI of $x\sin 3x$, one answer ESA ★

ADDED FOR CLARITY S006·4 · CW U2 P19

## Type 5 formula or complex exponential?

| | **Type 5 formula (slide)** | **Im/Re of $xe^{iax}$** |
|---|---|---|
| **Steps** | two Type 2 computations | one shift, one Type 3 series with complex numbers |
| **Needs** | $f'(D)$ and $(f(D))^2$ | $f(D+ia)$ and complex division |
| **Best for** | the exam (it is the taught rule) | checking the answer |
| **CW P19** | $-\frac{x\sin 3x}{10} - \frac{3\cos 3x}{50}$ | same |
| **CW P20** | $\frac{x(\cos 2x - 5\sin 2x)}{52} - \frac{73\sin 2x}{1352} - \frac{5\cos 2x}{169}$ | same |

ADDED FOR CLARITY S006·4 · CW U2 P19 · CW U2 P20